# View Reviews

**Paper ID**
2739

**Paper Title**
CATALYST: CATalogue-driven LLM sYstem for Spatial Tiling

**Track Name**
Demo

**Reviewer #1**

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## Questions

**1. Overall Rating**
Weak Reject

**2. The architecture and functionalities of the tool/software are clearly articulated.**
Yes

**3. The demonstration scenario is clearly articulated**
Yes

**4. Please rate the potential degree of audience interaction of the demo scenario**
Medium

**6. At most three strong points about the proposal. Please be precise; clearly explain the value and nature of the contribution.**
S1. CATALYST addresses three practical barriers in geospatial data utilization: discovery, scalable visualization, and styling, in one integrated system.

S2. The combination of semantic indexing for discovery with tile histograms for visualization is elegant. The dual approach of precomputing dense tiles while generating sparse tiles on-demand balances storage overhead with interactive latency effectively.

S3 The multi-turn refinement capability enables users to iteratively adjust both dataset selection and styling without restating context, making the system more usable for exploratory analysis. Multiple demonstration scenarios effectively showcase different interaction patterns.

**7. At most three weak points about the proposal. Please be precise here. The details can be in the relevant field below.**

W1. Claims about handling "hundreds of gigabytes" lack supporting experiments. No discussion of failure modes, data type limitations, or how performance degrades with dataset complexity.
W2. While acknowledged in Future Work, the security concerns of LLM-generated code execution should be addressed before production deployment.

**8. Elaborate on the above fields. Please number each point and provide as constructive feedback as possible.**

D1. The paper doesn't explain how semantic retrieval handles domain-specific terminology, how incomplete metadata is handled, or what sampling strategy is used for tile size limits. Scalability claims are not backed by testing. There's no discussion of how the system performs with ambiguous queries or how visualization accuracy is affected by sampling.
D2. Running LLM-generated JavaScript code presents security risks. The authors do acknowledge this issue for future work and have proposed interesting steps for mitigating it. Nonetheless, addressing these concerns should take precedence over the (equally interesting) cross-dataset exploration.
D3. No clear research contribution. It combines a pipeline of techniques (semantic search, tile-based mapping, LLM styling) in a practical way. The contributions and novel application of this tool could be highlighted more.

**Reviewer #2**

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## Questions

**1. Overall Rating**

Reject

**2. The architecture and functionalities of the tool/software are clearly articulated.**

Partially

**3. The demonstration scenario is clearly articulated**

Partially

**4. Please rate the potential degree of audience interaction of the demo scenario**
Low

**6. At most three strong points about the proposal. Please be precise; clearly explain the value and nature of the contribution.**
S1. The semantic exploration of datasets and their styled visualization is an interesting topic

**7. At most three weak points about the proposal. Please be precise here. The details can be in the relevant field below.**
W1. The demo appears to repeat previous demonstrations with only a few incremental additions that are not adequately demonstrated.
W2. The demonstration scenario is not well designed, requiring users to upload datasets.
W3. Both the video and the paper examples stick to very small datasets, that can be pre-processed and visualized fast, raising concerns about performance if a user picks a larger or more complex dataset.

**8. Elaborate on the above fields. Please number each point and provide as constructive feedback as possible.**
D1. This demo appears to share a very large amount of common features with the LASEK [2] demo, with the main new addition being that it has one more semantic search level across datasets to pick the data to visualize from. All the automatic and interactive styling and attribute search features seem to have already been demonstrated already with LASEK. To make the new version stand out, the authors should consider how would the tool behave in cases where there is relevant data in more datasets than just the top-1 most relevant. For example, the Riverside vegetation dataset is selected when a user asks about vegetation distribution types, but the second most relevant dataset being the Parks dataset, could very well include such data as well (even though it probably doesn't). What would happen in that case? This aggregation of information would really solidify and distinguish CATALYST against LASEK, making for a more interesting demo as well.

D2. The main interactive feature of the demo is that users are required to upload their own datasets, but it is unclear how they are expected to do this in practice. Do they select from a set of datasets provided by the authors, or are they expected to search online for openly available data? This introduces several challenges, as users may not be familiar with suitable data sources or may spend considerable time searching for interesting datasets. There are also practical limitations such as download time, and there is no guarantee that the data will be in the expected format, free of junk or null values, or even that it will include enough metadata. Hence and going back to the the points I made in D1, focusing on cross-dataset semantic search and information aggregation would make the tool easier to demonstrate and use. In such a setup, the authors would only have to provide a collection of pre-downloaded and pre-processed datasets, while the users could focus on exploring and formulating interesting queries across them in a combined manner.

D3. The tool reports semantic search time, LLM request time, and total request time, which provides some insight into its performance. However, (a) pre-processing time is not reported at all and (b) all datasets used appear to be unrealistically small. This raises concerns about the tool's feasibility in terms of performance and potential waiting time when the dataset collection becomes larger (which is the primary target use case for such a tool) and when individual datasets grow bigger (as it is usually the case with real-world spatial datasets).

Generally, in my opinion, for this demo to really distinguish itself from its earlier version (LASEK), it should focus more on cross-dataset search and the aggregation of all relevant information to be visualized, as well as a more interactive way to allow users to test the tool live during a demonstration than having to upload their own datasets.

**Reviewer #3**

## Questions

**1. Overall Rating**

Reject

**2. The architecture and functionalities of the tool/software are clearly articulated.**

Partially

**3. The demonstration scenario is clearly articulated**

Partially

**4. Please rate the potential degree of audience interaction of the demo scenario**

Medium

**6. At most three strong points about the proposal. Please be precise; clearly explain the value and nature of the contribution.**

+ The system provides a seamless and automated workflow, successfully integrating raw geospatial data upload, semantic metadata enrichment, and LLM-driven visual exploration.
+ The visualization index intelligently balances storage costs and rendering latency by precomputing dense vector tiles offline while generating sparse tiles dynamically.

**7. At most three weak points about the proposal. Please be precise here. The details can be in the relevant field below.**

- The LLM's functionality appears limited to a natural language-to-code translator based on attribute names, without demonstrating deep understanding of complex spatial topologies or geospatial reasoning.
- Directly executing LLM-generated JavaScript within the client's map runtime without any mentioned sandboxing or validation mechanisms poses significant security and system reliability risks.

**8. Elaborate on the above fields. Please number each point and provide as constructive feedback as possible.**

1. To justify the claim of being an "interactive" system, the authors must provide a quantitative breakdown of the system's latency. Please include the time consumed by semantic retrieval, LLM prompt generation/inference, and the final MVT rendering process.

2. The current scenarios seem heavily focused on single-dataset styling. Real-world spatial analytics often require overlaying or joining multiple datasets (e.g., visualizing wildfire hotspots over population density maps). The authors should acknowledge this limitation and discuss how the catalogue index might support cross-dataset queries in the future.